

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
Department of Computer Science and Engineering

**ASSESSMENT TOOL 1 – C Code Output Prediction**

|                  |                                                                                                               |               |                                              |
|------------------|---------------------------------------------------------------------------------------------------------------|---------------|----------------------------------------------|
| Course Code:     | CSA0305                                                                                                       | Course Title: | Data Structures                              |
| CO Assessed:     | CO1 – Imitation (BL3, BL4)                                                                                    | Assessment:   | Assessment Tool 1 – C Code Output Prediction |
| Weightage:       | 35% of CIA                                                                                                    | Total Marks:  | 100                                          |
| CO1 Statement:   | Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. |               |                                              |
| Student Name:    | K.Navya                                                                                                       | Reg. No.:     | 192511231                                    |
| Section / Batch: | 2 <sup>nd</sup> Year CSE                                                                                      | Date:         | 08/07/2026                                   |

**Code of Conduct**

*I K.Navya(1925112311)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: K.Navya

**Instructions**

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20\*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

| Section / Topic                                                                   | Focus                                      | Q Nos  | Marks     |
|-----------------------------------------------------------------------------------|--------------------------------------------|--------|-----------|
| Linked data structure                                                             | Basic data structure                       | 1 - 2  | 20        |
| Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure | Classifications of Linear data structure   | 3 - 4  | 20        |
| Search Data Structure                                                             | Linear Search and Binary Search            | 5 -6   | 20        |
| Recursion, Performance analysis                                                   | Recursion                                  | 7 - 8  | 20        |
| Array Data Structures                                                             | Implementations Using Array data structure | 9 - 10 | 20        |
| GRAND TOTAL:                                                                      |                                            |        | 100 Marks |

**CO1\_AT1\_C CODE OUTPUT PREDICTION**

← Back

**QUESTIONS**
**Q1**  
C Code - Pointer with Array  
10 marks

**Q2**  
C Code - Linked List Node  
10 marks

**Q3**  
C Code - Binary Search  
Complexity  
10 marks

**Q4**  
C Code - Pointer Increment  
10 marks

**Q5**  
C Code - Array Address  
10 marks

**Q6**  
C Code - Linked List Traversal  
10 marks

**Q1 C Code - Pointer with Array**

10 marks

```
#include<stdio.h>
int main()
{
    int arr[]={10,20,30};
    int *p=arr;
    printf("%d",*(p+2));
}
```

**Your Answer**

30

Save Answer

**CO1\_AT1\_C CODE OUTPUT PREDICTION**

← Back

**QUESTIONS**
**Q1**  
C Code - Pointer with Array  
10 marks

**Q2**  
C Code - Linked List Node  
10 marks

**Q3**  
C Code - Binary Search  
Complexity  
10 marks

**Q4**  
C Code - Pointer Increment  
10 marks

**Q5**  
C Code - Array Address  
10 marks

**Q6**  
C Code - Linked List Traversal  
10 marks

**Q2 C Code - Linked List Node**

10 marks

```
#include<stdio.h>
struct Node{
    int data;
    struct Node *next;
};
int main(){
    struct Node n1,n2;
    n1.data=10;
    n2.data=20;
    n1.next=&n2;
    n2.next=NULL;
    printf("%d",n1.next->data);
}
```

**Your Answer**

20

**CO1\_AT1\_C CODE OUTPUT PREDICTION**

← Back

**QUESTIONS**
**Q1**  
C Code - Pointer with Array  
10 marks

**Q2**  
C Code - Linked List Node  
10 marks

**Q3**  
C Code - Binary Search Complexity  
10 marks

**Q4**  
C Code - Pointer Increment  
10 marks

**Q5**  
C Code - Array Address  
10 marks

**Q6**  
C Code - Linked List Traversal  
10 marks

**Q3 C Code - Binary Search Complexity**

10 marks

```
while(low <= high){
    mid = (low + high) / 2;
}
```

**Your Answer**

O(log n)

Save Answer

**CO1\_AT1\_C CODE OUTPUT PREDICTION**

← Back

**QUESTIONS**
**Q1**  
C Code - Pointer with Array  
10 marks

**Q2**  
C Code - Linked List Node  
10 marks

**Q3**  
C Code - Binary Search Complexity  
10 marks

**Q4**  
C Code - Pointer Increment  
10 marks

**Q5**  
C Code - Array Address  
10 marks

**Q6**  
C Code - Linked List Traversal  
10 marks

**Q4 C Code - Pointer Increment**

10 marks

```
#include <stdio.h>
int main() {
    int a[] = {5, 10, 15};
    int *p = a;
    p++;
    printf("%d", *p);
}
```

**Your Answer**

10

Save Answer

**CO1\_AT1\_C CODE OUTPUT PREDICTION**
[< Back](#)
**QUESTIONS**
**Q1**  
C Code - Pointer with Array  
10 marks

**Q2**  
C Code - Linked List Node  
10 marks

**Q3**  
C Code - Binary Search  
Complexity  
10 marks

**Q4**  
C Code - Pointer Increment  
10 marks

**Q5**  
C Code - Array Address  
10 marks

**Q6**  
C Code - Linked List Traversal  
10 marks

**Q5 C Code - Array Address**

10 marks

```
int arr[]={5,10,15};
printf("%d",*(arr+1));
```

**Your Answer**

10

 Save Answer

**CO1\_AT1\_C CODE OUTPUT PREDICTION**
[< Back](#)
**QUESTIONS**
**Q1**  
C Code - Pointer with Array  
10 marks

**Q2**  
C Code - Linked List Node  
10 marks

**Q3**  
C Code - Binary Search  
Complexity  
10 marks

**Q4**  
C Code - Pointer Increment  
10 marks

**Q5**  
C Code - Array Address  
10 marks

**Q6**  
C Code - Linked List Traversal  
10 marks

**Q6 C Code - Linked List Traversal**

10 marks

```
Node1 -> Node2 -> Node3 (Node values are 10,20,30)
current=head;
current=current->next;
printf("%d",current->data);
```

**Your Answer**

20

 Save Answer

**CO1\_AT1\_C CODE OUTPUT PREDICTION**
[< Back](#)
**QUESTIONS**
**Q1**  
C Code - Pointer with Array  
10 marks

**Q2**  
C Code - Linked List Node  
10 marks

**Q3**  
C Code - Binary Search  
Complexity  
10 marks

**Q4**  
C Code - Pointer Increment  
10 marks

**Q5**  
C Code - Array Address  
10 marks

**Q7 C Code - Recursion (Factorial)**

10 marks

```
#include <stdio.h>
int fact(int n){
    if(n==0)
        return 1;
    return n * fact(n-1);
}
int main(){
    printf("%d", fact(4));
    return 0;
}
```

**Your Answer**

24

**CO1\_AT1\_C CODE OUTPUT PREDICTION**

← Back

**QUESTIONS**
**Q1**  
C Code - Pointer with Array  
10 marks

**Q2**  
C Code - Linked List Node  
10 marks

**Q3**  
C Code - Binary Search  
Complexity  
10 marks

**Q4**  
C Code - Pointer Increment  
10 marks

**Q5**  
C Code - Array Address  
10 marks

**Q8 C Code - Recursion (Sum)**

10 marks

```
#include <stdio.h>
int sum(int n){
    if(n==0)
        return 0;
    return n + sum(n-1);
}
int main(){
    printf("%d", sum(5));
}
```

Your Answer

15

**CO1\_AT1\_C CODE OUTPUT PREDICTION**

← Back

**QUESTIONS**
**Q1**  
C Code - Pointer with Array  
10 marks

**Q2**  
C Code - Linked List Node  
10 marks

**Q3**  
C Code - Binary Search  
Complexity  
10 marks

**Q4**  
C Code - Pointer Increment  
10 marks

**Q5**  
C Code - Array Address  
10 marks

**Q9 C Code - Primitive Data Type**

10 marks

```
#include <stdio.h>
int main(){
    int a=10;
    char ch='A';
    printf("%d %c",a,ch);
}
```

Your Answer

10 A

**CO1\_AT1\_C CODE OUTPUT PREDICTION**

← Back

**QUESTIONS**
**Q1**  
C Code - Pointer with Array  
10 marks

**Q2**  
C Code - Linked List Node  
10 marks

**Q3**  
C Code - Binary Search  
Complexity  
10 marks

**Q4**  
C Code - Pointer Increment  
10 marks

**Q5**  
C Code - Array Address  
10 marks

**Q6**  
C Code - Linked List Traversal  
10 marks

**Q10 C Code - Linear Data Structure (Array)**

10 marks

```
#include <stdio.h>
int main(){
    int arr[]={2,4,6,8};
    printf("%d",arr[3]);
}
```

Your Answer

8

Save Answer